

Curve Fitting

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 3

Learning Outcomes

- Associate curves with mathematical functions.
- Select suitable functions to fit to data.
- Distinguish between interpolation and extrapolation.
- Use MATLAB's `cftool` to fit curves to data.
- Assess the quality of a fitted model.

Curve Fitting

Scientists and engineers frequently collect experimental and observational data. Often the underlying mathematical relationship between two variables is not known.

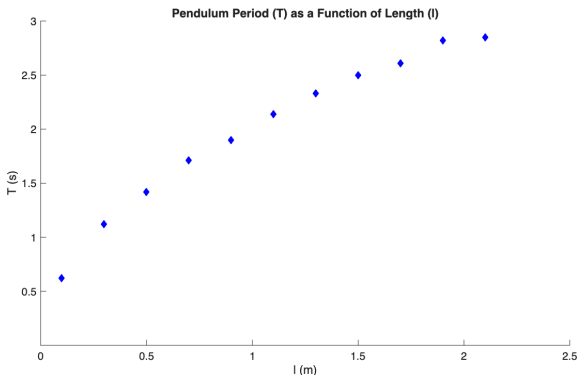
Curve fitting is the process of finding a mathematical function that closely describes the observed data. It tells us nothing about why a system behaves in a certain way, but can be an important step towards finding that out.

A fitted curve can then be used to:

- describe patterns,
- estimate missing values,
- make predictions.

The General Problem

Suppose you have collected some bivariate (involving two variables) data about a system, for example, the period of a pendulum for different lengths. Plotting the data in a scatter graph we get



The General Problem

The question we wish to answer is:

“What function, when plotted, closely resembles the data?”

Since we are attempting to fit a mathematical curve to measured data, the process is called **curve fitting**.

Linear and Quadratic Functions

Linear

$$y = a_0 + a_1x$$

- Straight line
- Constant gradient
- Constant rate of change

Quadratic

$$y = a_0 + a_1x + a_2x^2$$

- Curved trend
- One turning point
- Parabolic shape

Exponential and Logarithmic Functions

Exponential

$$y = Ae^{Bx} + C$$

- Rapid growth or decay
- Common in natural processes

Logarithmic

$$y = A \ln(Bx) + C$$

- Rapid initial change
- Growth slows over time

Power Laws and Sinusoids

Power

$$y = Ax^n + B$$

- Scaling relationships
- Common in engineering

Sinusoidal

$$y = A \sin(\omega x + \phi) + B$$

- Repeating behaviour
- Oscillations and vibrations

Which Function Would You Choose?

For each graph, identify the most suitable function family.

Function	Name
$y = a_0 + a_1x$	Linear
$y = a_0 + a_1x + a_2x^2$	Quadratic
$y = a_0 + a_1x + a_2x^2 + a_3x^3$	cubic
$y = \frac{A}{x} + B$	Hyperbolic
$y = Ax^n + B$	Power law
$y = A \sin(\omega x + \phi) + B$	Sinusoid
$y = A \ln(Bx) + C$	Logarithmic
$y = Ae^{Bx} + C$	Exponential

Interpolation and Extrapolation

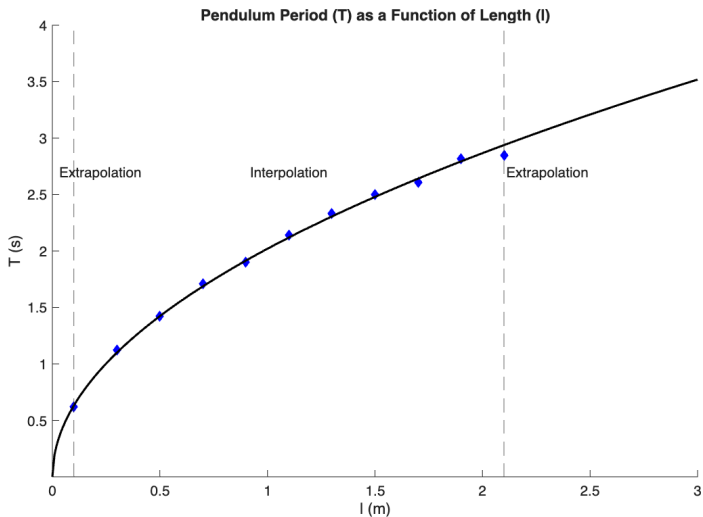
Once a curve has been fitted, it can be used to estimate unknown values.

Interpolation Estimating values within the range of the original data.

Extrapolation Estimating values outside the range of the original data.

Extrapolation is generally less reliable because future behaviour may differ from the observed trend.

Interpolation and Extrapolation



Trial and Error

A crude curve fit can be achieved as follows:

- Select an appropriate function to match the pattern of the data.
- Choose some broadly sensible values for the parameters, e.g. a_0 and a_1 in $y = a_0 + a_1x$.
- Use MATLAB to plot your curve in the same graph as your data and adjust the parameters until the curve matches the data as best it can.

Trial and Error

However this approach has several limitations.

- Slow.
- Subjective.
- Difficult to reproduce.
- Often less accurate.

Modern engineering applications typically use specialised software.

MATLAB cftool

MATLAB includes a dedicated curve fitting application called `cftool`.

The user may:

- choose a predefined model;
- create a custom model.

MATLAB automatically determines the parameter values that produce the best fit.

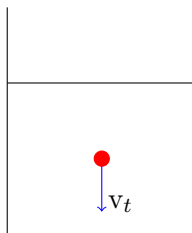
Curve Fitting Example

An experimenter wants to know if an object's size will have any influence on its terminal velocity when it is dropped through a viscous fluid, such as oil. So, they drop spheres of different diameter d through some oil and record the terminal velocity v_t in each case.

Establish the mathematical relationship connecting v_t and d by fitting a curve to the data.

Experimental Data

d (mm)	v_t (mm/s)
2	13.7
4	54.8
8	219.2
10	342.6
15	770.8
20	1370.3



Before We Fit

Notice the first few measurements:

$$2 \rightarrow 13.7$$

$$4 \rightarrow 54.8$$

$$8 \rightarrow 219.2$$

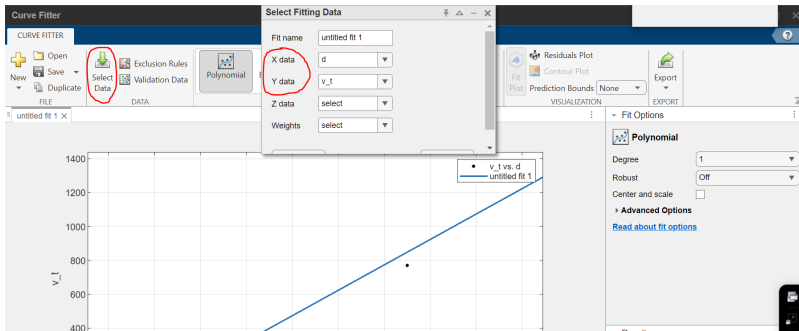
When diameter doubles, terminal velocity approximately increases by a factor of four.

This suggests a possible power-law relationship.

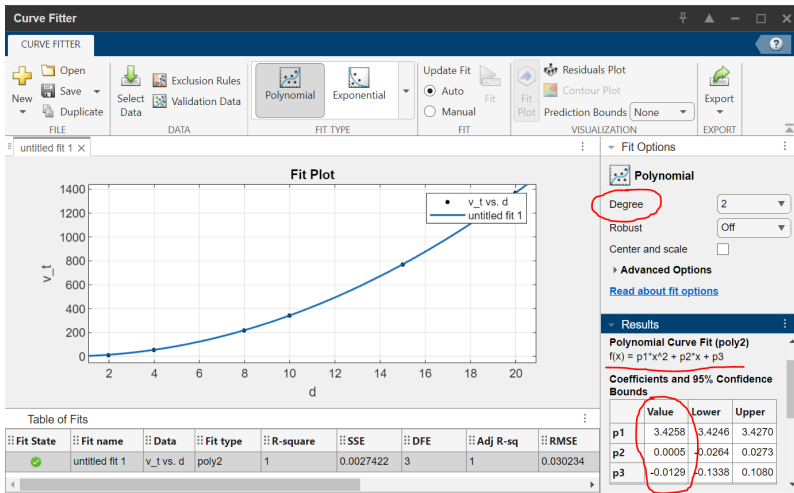
Using the cftool

```
1 clear all
2 T = readtable("Lecture_4_cftool.xlsx"); % Reads in data from a spreadsheet called *.xlsx
3 data = table2array(T); % Puts data into a matrix called 'data'
4 d = data(:,1); % horizontal axis values are being taken from column 1
5 v_t = data(:,2); % vertical axis values are being taken from column 2
6 %cftool
```

Using the cftool



Using the cftool



Goodness of Fit

A fitted curve should represent the data accurately. This is known as the **goodness of fit**.

One common measure is the coefficient of determination:

$$R^2$$

which takes values between 0 and 1.

Goodness of Fit and the Role of R^2

In general, the following ranges are often used to judge the quality of a fit:

- $R^2 > 0.9$ indicates an excellent fit
- $0.7 < R^2 < 0.9$ indicates a good fit
- $0.5 < R^2 < 0.7$ indicates a moderate fit
- $R^2 < 0.5$ indicates a weak fit

It is important to remember, however, that a high R^2 value does not always mean the model is the most appropriate choice. For example, overly complex models can give artificially high R^2 values by “overfitting” the data.

Root Mean Square Error (RMSE)

Another common measure of fit quality is the **Root Mean Square Error (RMSE)**.

RMSE measures the average size of the errors between the measured data and the fitted curve.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where

- y_i is the measured value;
- $\hat{y}_i$ is the predicted value from the model;
- n is the number of data points.

Interpreting RMSE

The RMSE has the same units as the measured data.

For example, if the velocity is measured in mm/s then the RMSE is also measured in mm/s.

- Small RMSE $\Rightarrow$ predictions are close to the data.
- Large RMSE $\Rightarrow$ predictions differ significantly from the data.
- $\text{RMSE} = 0$ corresponds to a perfect fit.

When comparing models fitted to the same data set, a smaller RMSE generally indicates a better fit.

RMSE and R^2

MATLAB's `cftool` typically reports both RMSE and R^2 .

- RMSE measures the typical size of the prediction error.
- R^2 measures how much of the variation in the data is explained by the model.

A good model usually has:

- a high value of R^2 ;
- a low value of RMSE.

Neither statistic should be used on its own. The fitted curve should always be inspected visually.